

Measuring the dip of the sea horizon as a function of altitude

A high signal-to-noise test, feasible with a camera, a tray of liquid and a star, whose two possible outcomes are stated in advance.

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ABSTRACT

Horizon dip — the angle δ between the local horizontal, defined by the plumb line, and the sea horizon — is a geometric observable about which two models of the Earth's surface make incompatible, non-overlapping predictions. A spherical model predicts $\delta = \sqrt{(2h/R')}$, growth as the square root of altitude; a plane model predicts $\delta = 0$ at every altitude, the horizon being the vanishing line of the plane and therefore lying, by construction, at eye level.

The metrological value of this observable lies in the fact that atmospheric refraction — which dominates the uncertainty of most long-range geodetic measurements — enters only at second order: across the entire physically defensible range of the refraction coefficient ($0 \leq k \leq 0.47$), the spherical prediction varies only from 107' to 78' at 3 107 m elevation. **The gap between the two models is therefore at least 78 arcminutes, against an estimated instrumental error budget of 2.2' (1 σ) — a signal-to-noise ratio of order 36 against the plane hypothesis, and of 29 on the determination of the coefficient a , where refraction uncertainty enters in full.** The two ratios are distinguished in section 08.

We propose a distributed campaign: measure δ from a range of altitudes spanning a few metres to a few kilometres, and test not an isolated point but the **law** $\delta(') = a \cdot \sqrt{h(m)}$. The spherical model requires $1.40 \leq a \leq 1.93$ depending on refraction; the plane model requires $a = 0$. Decision criteria are fixed in section 07, prior to any acquisition.

01 Purpose

This document invites independent observers to carry out a simple angular measurement whose result discriminates between two geometric models of the Earth's surface, and to publish their raw data whatever the outcome.

The choice of observable is not arbitrary. Most observations invoked in this debate — visibility of distant objects, long-range photography, levelling of water surfaces — share a common weakness: their interpretation depends on a poorly constrained atmospheric parameter, the refraction coefficient k , whose real variability is sufficient to span the gap between the models. The argument then shifts onto the value of k and never concludes.

Horizon dip escapes this trap for a structural reason: it varies as $\sqrt{(1-k)}$, that is, very slowly, whereas hidden-height measurements vary as $1/(1-k)$ and as the square of distance. Multiplying k fourfold reduces δ by only 27 per cent. No value of k brings a sphere to $\delta = 0$.

02 Theoretical basis

For an observer whose eye is at height h above a spherical surface of radius R , the grazing line of sight touches the surface at the point of tangency. The angle between that line and the local horizontal plane is, to first order in h/R :

$$\delta = \sqrt{(2h / R')}, \quad R' = R / (1 - k)$$

δ in radians, h and R' in metres. k = terrestrial refraction coefficient.

Expressed in arcminutes with h in metres, the relation becomes a single-parameter square root law:

$$\delta(') = a \cdot \sqrt{h(\text{m})}, \quad a = 3\,437.75 \cdot \sqrt{(2(1-k) / R)}$$

The coefficient a is the only quantity to be measured. It is the slope of δ against $\sqrt{h}$.

On a plane surface, the horizon is the vanishing line of the surface plane. In perspective projection a vanishing line lies exactly in the plane through the eye parallel to the receding plane — here, at eye level. The prediction is therefore $\delta = 0$, independent of h : this is a property of projective geometry, not an approximation.

REGIME	k	$a \text{ (}' \cdot \text{m}^{-1/2})$	$\delta \text{ at } 3\,107 \text{ m}$
Optical vacuum (theoretical bound)	0.00	1.9261	107.4'
Free atmosphere, high altitude	0.10	1.8273	101.9'
Geodetic standard	0.13	1.7966	100.1'
Cold sea, moderate inversion	0.17	1.7548	97.8'
Super-refraction	0.25	1.6681	93.0'
Atmospheric duct, Fata Morgana	0.47	1.4022	78.2'
Plane model — any condition	—	0.0000	0.0'

Table 1 — Coefficient a of the square root law, by refraction regime.

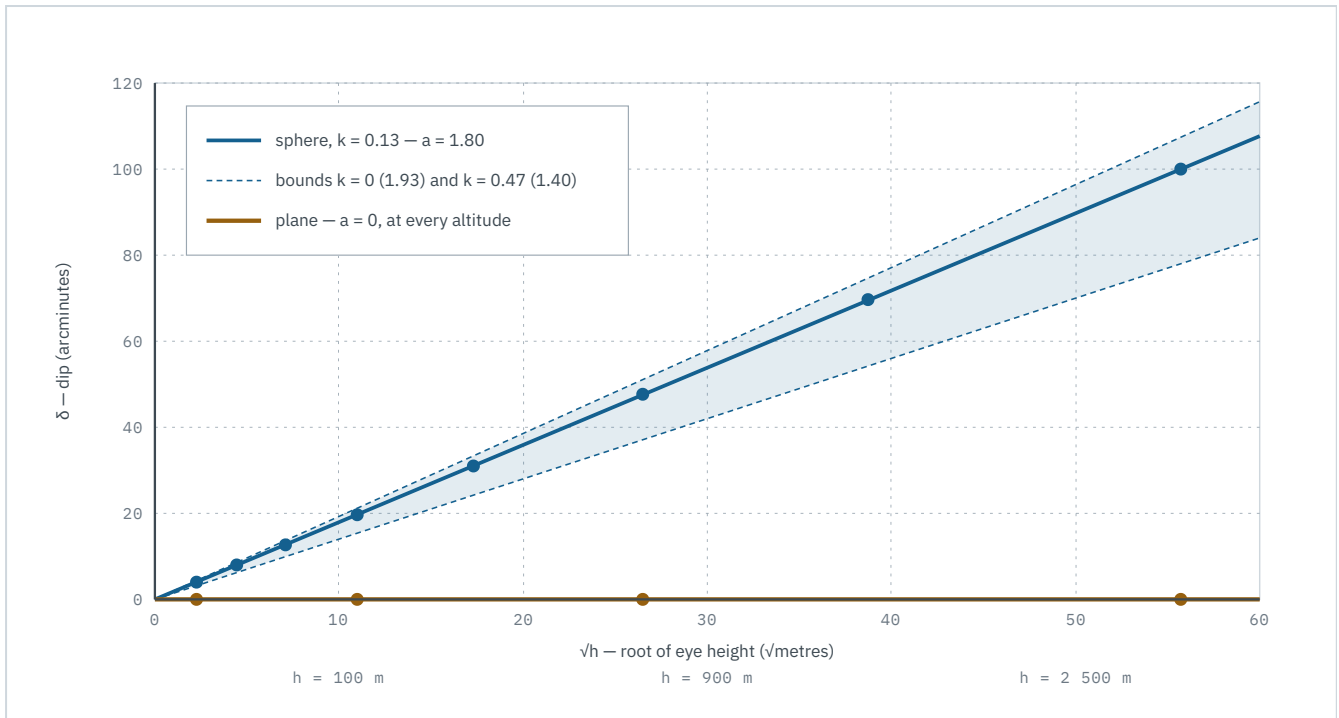


Figure 1 — The observable, and why it is a curve rather than a point. Plotting δ against $\sqrt{h}$ linearises the spherical prediction into a straight line through the origin. The blue band covers *the whole* physically defensible refraction range, $k = 0$ to $k = 0.47$; its width is the model's irreducible uncertainty. The ochre line is the plane model's prediction. The two domains are disjoint above $h \approx 3$ m. A campaign at several altitudes tests not a value but **a law**: slope, linearity and passage through the origin are three independent constraints.

If refraction is granted to the plane model

The objection is legitimate and must be answered: the spherical prediction includes refraction, the plane's is allowed only straight-line propagation. The comparison looks unbalanced.

It is not, and the imbalance favours the plane model. Ray curvature is $1/\rho$ with $k = R/\rho$: it depends only on the atmosphere, and is therefore the same above a plane. The *relative* curvature of ray and surface then becomes:

$$\text{sphere: } 1/R - 1/\rho = (1-k)/R \quad | \quad \text{plane: } 0 - 1/\rho = -k/R$$

Positive on a sphere while $k < 1$: a horizon exists. Negative on a plane for any k .

On a plane with a real atmosphere the ray therefore curves **more** than the surface: the ground rises around the observer instead of falling away. From 3 107 m, sighting 1° above the horizontal, the ground would be met at 1 873 km; at 5° , at 8 589 km. **No horizon would exist at all** and the ground would fill the sky.

WHAT THIS ESTABLISHES

$\delta = 0$ is the plane model's **best case**, obtained by granting it straight-line propagation. Granting it refraction does not improve matters: it leads to an absurd and massively refuted prediction. This is also why no intermediate value is available: the plane model has no mechanism producing a dip of a few tens of arcminutes.

Exact form

The exact relation is $\delta = \arccos(R' / (R' + h))$, of which $\sqrt{(2h/R')}$ is the first-order expansion. The difference is $0.018'$ at 3 107 m and $0.036'$ at 5 000 m — below the error budget, but the exact form costs nothing and is the one to use in reduction.

03 Pre-registered predictions

Both hypotheses are stated here in falsifiable form, before any data are acquired. Neither contains a free parameter beyond k , which is itself bounded.

HYPOTHESIS S — SPHERICAL SURFACE

$\delta = a\sqrt{h}$, $1.40 \leq a \leq 1.93$

The line passes through the origin. Scatter of the points about the fitted line measures refraction variability across sites and epochs.

HYPOTHESIS P — PLANE SURFACE

$\delta = 0$ for all h

No dependence on altitude. Any measured departure is attributable to instrumental error and must average to zero.

TYPICAL SITE	h (m)	$\sqrt{h}$	δ — hyp. S ($k=0.13$)	δ — hyp. S ($k=0$)	δ — hyp. P
Beach, jetty	5	2.24	4.0'	4.3'	0'
Sea wall, ship's deck	20	4.47	8.0'	8.6'	0'
Low cliff	50	7.07	12.7'	13.6'	0'
Lighthouse, tower block	120	10.95	19.7'	21.1'	0'
Coastal hill	300	17.32	31.1'	33.4'	0'
Mountain road	700	26.46	47.5'	51.0'	0'
Medium summit	1 500	38.73	69.6'	74.6'	0'
High altitude	3 107	55.74	100.1'	107.4'	0'

Table 2 — Expected values across a range of accessible altitudes. The two right-hand columns do not overlap at any altitude above 3 m.

04 Order of operations

Every necessary element is present in the sections that follow, but scattered across six of them. And the order is not indifferent: several steps make the preceding ones unrecoverable. This section does not repeat the procedures — it says in what order to take them, and which one cannot be redone.

THE FIVE THINGS THAT CAN NEVER BE RECOVERED AFTERWARDS

1. **The timestamp.** Deposited after the first frame, it proves nothing. Nothing that follows has evidential value without it.
2. **Cartographic checking of the azimuth.** A distant coastline in the sightline is indistinguishable from the horizon in the photograph. The reading is lost and it looks sound.
3. **The elevation above the sea at the time.** Below 50 m it requires an on-site measurement. Home again with a GPS height, you no longer have a station.
4. **ΔT , tide and sea state.** Recorded at the moment of exposure, or never. Tide can be recovered from tables; sea state and water temperature cannot.
5. **Taping the focus ring.** It must precede the measurement frame. If focus shifts between that frame and the solving frame, the scale no longer carries across and the measurement has no unit.

Phase 0 — before the first frame, once for the whole campaign

01 Deposit the protocol

Freeze the PDF, compute its SHA-256 hash, deposit the file on OSF or Zenodo, obtain the DOI, publish DOI and hash. Section 12. **Until that is done, photograph nothing.**

02 Fix the pointing criterion

The one for the horizon line, stated in section 06, and circulate it to every observer. It matters less that the criterion be the best than that it be *the same everywhere*: a bias common to all shows up in the fitted intercept, a bias that varies between observers shows up nowhere.

03 Choose the spread of stations

At least one above 300 m — below that the test is not decisive, as table 11 quantifies. Spread the others evenly in $\sqrt{h}$, not in h : $\sqrt{h}$ is the fitting variable.

04 Draw the order

If the fallback is used, the normal/inverted alternation is drawn before the session, not during it.

Phase 1 — preparing the station, several days ahead

01 Check the azimuth on a chart

Open sea over the whole theoretical range for the altitude — 213 km from 3 107 m, 12 km from 10 m. No coastline, no island, no shoal in the sightline. **Irreversible step:** neglected, it yields a normal-looking reading of no value.

02 Establish the eye's elevation

Above the *sea surface at the time*, orthometric. Not the GPS ellipsoidal height. Below 50 m, plan a direct on-site measurement to the decimetre. Section 08.

03 Locate the window

Ephemerides: solar altitude, target, its azimuth and true altitude. Venus or Jupiter in preference to a star — they can be acquired well before nightfall, when the sea horizon is still fully legible. Section 06.

04 Get the tide time

For the chosen slot, and if possible prefer a low-range moment: at 5 m elevation, two metres of tide are worth 41 per cent of δ .

Phase 2 — on site, before the frames. Order matters

01 Safety first

This protocol sends people onto cliffs and summits, at night. Scout the ground by day, do not work alone at the edge, and plan for red light and cold. No measurement is worth an accident.

02 Set up the tripod and the optics

Focus at infinity, intermediate aperture. Then **tape the focus ring** and do not touch it again until the night is over — the solving frame depends on it.

03 Set up the tray

Off the tripod, on sand or foam, sheltered from wind, viscous liquid, read at the centre. Then **let it settle**: it is the only moment of the evening when waiting is productive.

04 Record the conditions

Air temperature at eye height, sea surface temperature, ΔT , pressure, humidity, wind, sea state and significant wave height. **Irreversible step.**

05 Rehearse the framing

Target, reflection and horizon line in the field, *before* the window opens and not during it. The window lasts twenty to thirty minutes; it is not the moment to discover that two degrees of field are missing.

Phase 3 — acquisition

01 The burst

Twenty successive frames on the same sightline. Raw file, uncropped. An isolated frame has no estimable uncertainty and does not count as a measurement.

02 Repeat

Three series minimum, at distinct hours, to sample refraction variability.

03 Cross the methods

If both rigs are available: one series with the artificial horizon and one with the inclinometer and reversal, same site, same hour. This cannot be made up on another date.

04 The solving frame

In full darkness, same setting, ring still taped: a populated star field, meant solely for astrometry. It is the last frame of the evening, and forgetting it costs the whole evening.

Phase 4 — reduction, at the desk, in this order

01 Solve the field

Scale in arcseconds per pixel, residuals, distortion profile. Everything else is expressed in that unit: this is where to start.

02 Pass the free check

Target-to-reflection separation divided by two, compared with the ephemeris altitude plus refraction. **If they do not agree to 0.5 per cent, stop and find out why** before reducing anything. Section 06.

03 Point the horizon

By the criterion fixed in phase 0, without exception, recording the column-to-column standard deviation and the number of columns discarded.

04 Reduce δ

The midpoint between target and reflection gives the true horizontal; the offset between that midpoint and the horizon line is δ .

05 Measure the uncertainties, do not postulate them

Pointing σ = scatter of the horizon position over the burst. Surface σ = scatter of the reflection over the same burst, divided by two.

06 Correct the elevation

For the actual tide at the moment of exposure, not the mean tide.

07 Submit

The full record sheet of section 11 and the raw files, **whatever the result**. A station is discarded only on a written and published ground; never because the number displeases.

Phase 5 — campaign, once every station is in

01 Linear fit

Weighted regression of δ against $\sqrt{h}$. Slope a , intercept b , and confidence intervals on both.

02 Free-exponent fit

$\ln \delta$ against $\ln h$. The exponent p must come out at 0.500, and its error bar says so.

03 Publish

Raw data, fits, and the conclusion they impose — including the stations in anomalous refractive regime, kept and flagged.

What is really needed

The next section details the equipment. Here is what actually has to be gathered, without frills — the whole set fits within a budget of second-hand gear and free software.

item	what actually serves
Camera	any interchangeable-lens body that writes raw. A second-hand DSLR does
Lens	a 100 mm prime. Avoid a zoom if you can; if you cannot, tape the ring
Tripod	stable. For the fallback only, it must allow inverted mounting
Reflecting surface	a wide, shallow baking dish, and pharmacy liquid paraffin or glycerine — a few decilitres
Something to stand it on	a box of sand, or a slab of foam. Never the tripod
Thermometers	an infrared thermometer aimed at the water, and an ordinary air thermometer
Barometer	a recent phone's will do, reset against the nearest weather station
Ephemerides	Stellarium, free and open
Angular scale	Astrometry.net, free and online; or ASTAP and SIRIL, free and offline
Tides	the SHOM tables, or the equivalent national service

Table 3 — What is needed, and what will stand in for it.

What to avoid absolutely

Two lists, and the distinction between them is worth keeping: the first destroys the measurement without leaving a trace, the second degrades it visibly.

WHAT DESTROYS THE MEASUREMENT

And does not show afterwards

Cropping the file — the optical centre is lost, and with it the reference.

Letting the raw converter correct the lens or lift contrast — the geometry changes.

Refocusing between the measurement frame and the solving frame — the scale no longer carries.

Taking the elevation from GPS without orthometric conversion — 178 per cent error at 10 m for a 30 m offset.

A coastline, an island or a haze bank in the sightline — that is not the geometric horizon.

A single frame — no estimable uncertainty.

Discarding a station because the number displeases.

WHAT DEGRADES THE MEASUREMENT

And can be put right by starting again

Water in the tray at an exposed site — use a viscous liquid.

The tray on the tripod — it passes on its vibration.

Reading near the edge of the tray — the meniscus curves the surface.

A glass cover without the 180° pair — its prism deflects the ray.

Trusting the engraved focal length (2–5 per cent) or the lunar standard (7 per cent).

An internal-focus lens, whose focal length varies with distance.

Working at one altitude only — it is a law being tested, not a point.

Confusing a level horizon in the frame with a level optical axis.

05 Equipment

Three lists: what serves in both cases, what method A requires as the reference procedure, and what the fallback requires. A single list built around the inclinometer would leave out the tray, the liquid and the ephemerides, without which method A cannot be carried out.

Common to both methods

- **Camera with sensor of known dimensions**, a prime or lockable lens, and a stable tripod. See the table below for choosing the focal length.
- **Unobstructed sea horizon** over at least the theoretical range for the altitude in question — 213 km from 3 107 m, 12 km from 10 m.
- **Barometer and thermometer**, to document conditions.
- **A second thermometer, for sea surface temperature** — a probe on a line, or an infrared thermometer aimed at the water. It is the air-to-water difference that governs the thermal gradient of the boundary layer, and therefore k at low stations. See section 09.
- **An elevation referred to the sea surface at the time**, not to the ellipsoid. This is the protocol's most dangerous point at low stations; see section 08.
- **Tide state and sea state** at the moment of acquisition.
- **An ephemeris application** — Stellarium, Sky Safari, or the IMCCE tables. It serves three purposes: choosing the target, knowing its true altitude to the arcsecond, and locating the twilight window.
- **Plate-solving software** — Astrometry.net online, ASTAP or SIRIL offline. It supplies the angular scale; see section 08.

For method A — artificial horizon, the reference procedure

- **A wide, shallow tray**. Thirty centimetres across is enough; what matters is the surface, not the volume.
- **A viscous liquid** — liquid paraffin or glycerine, a few decilitres. See the procedure for the reason.
- **Something to shelter the tray from wind**, and something to stand it on away from the tripod: a box of sand, a slab of foam.
- *Optional* — a flat glass cover, if wind demands it, with the 180° rotation procedure.

For the fallback — inclinometer

- **An inclinometer.** A digital construction level ($\pm 0.05^\circ$) suffices; a phone inclinometer ($\pm 0.5^\circ$) is acceptable provided reversal is applied.
- **A tripod allowing inverted mounting:** flip head, L-bracket, or reversed centre column.

focal length	vertical field	scale	admissible target altitude
50 mm	27.0°	0.41' /px	13.5°
85 mm	16.1°	0.24' /px	8.0°
100 mm	13.7°	0.21' /px	6.8°
135 mm	10.2°	0.15' /px	5.1°
200 mm	6.9°	0.10' /px	3.4°

Table 4 — Choosing the focal length, 24×36 sensor sampled at 4 000 pixels in height. The last column is the admissible target altitude: the field must hold the target at $+a$ and its reflection at $-a$, that is $2a$, and the dip itself must still fit, up to 1.7° .

The 100 mm remains the best compromise, and not merely for convenience: at 0.21' per pixel, three pixels of pointing uncertainty are worth 0.62', a quarter of the instrumental budget. A 50 mm would carry that same defect to 1.2', half the budget, for the sole advantage of a higher target. A 200 mm would do better still, but it admits only a target at 3.4°, where turbulence is worst.

06 Procedure — Method A, artificial horizon

This is the reference procedure; levelling remains the fallback. It **requires neither a levelling instrument nor a bias calibration**: the horizontal reference makes itself. One quantity remains to be established, the **angular scale**, and section 08 says how — the to speak of a procedure with “no calibration at all” would be one shortcut too many.

WHAT THE PROTOCOL ASSUMES HERE, AND WHAT IT DOES NOT

The protocol **advances no hypothesis about the cause** of the perpendicularity between a surface at rest and the plumb line. It uses only the *fact*, checkable with a set square by anyone, that it holds. The plumb line is an **instrument** here, not a theory: it is the material reference to which δ is referred, and the reasoning that follows is purely geometric.

A still water surface sets itself perpendicular to the plumb line. A star and its reflection are therefore symmetric about the true horizontal, and their midpoint materialises it. If the sea horizon line appears in the same frame, δ is read directly.

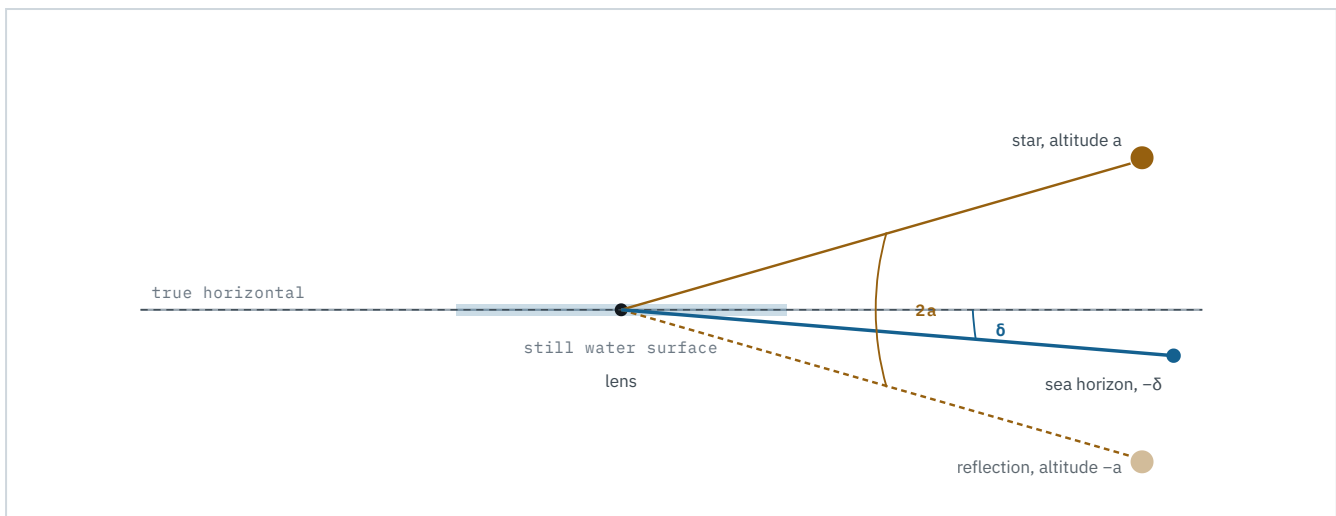


Figure 2 — The artificial horizon applied to dip. Star and reflection are symmetric about the true horizontal: their midpoint materialises it, with no levelling whatever. The sea horizon line, in the same frame, is then read directly at $-\delta$. Refraction cancels — direct and reflected images undergo it identically — and all three objects are at infinity, so a single focus setting serves.

01 Set up the surface

A wide, shallow tray, sheltered from wind. **Use a viscous liquid** rather than water: liquid paraffin ($\approx 70 \text{ mm}^2/\text{s}$) and glycerine ($\approx 1\,100 \text{ mm}^2/\text{s}$) damp capillary ripples tens to a thousand times faster than water ($1 \text{ mm}^2/\text{s}$). It is the only remedy for high-frequency vibration when a high-altitude site offers no genuinely calm shelter — and at 300 or 3 000 m that is the rule rather than the exception. Stand the tray on sand or foam, never on the tripod, which passes on its own vibration. Read at the centre of the tray, not near the edges where the meniscus curves the surface. Let it settle.

02 If wind forces a glass cover

Glass never has its two faces rigorously parallel, and that residual prism deflects the ray. Take a second frame with the cover turned through 180° and average the two: the prism cancels exactly, by the same reasoning as camera reversal.

03 Choose the target

Between 3° and 5° altitude, in the azimuth of the horizon under study. The vertical field required is $2a$, that is 6° to 10° ; a 100 mm lens offers 13.7° . **Aim for 3° rather than 5° :** at 5° the margin is only 1.85° on each side, and it must still absorb the dip, which reaches 1.7° at 3 107 m. A 100 mm is ample at 3° ; only just at 5° .

04 Frame all three

Star, reflection and horizon line in the same image. Focus at infinity: all three lie there, the reflection included.

05 Reduce

The midpoint between star and reflection gives the true horizontal. The offset between that midpoint and the horizon line is δ .

06 Repeat

Three series minimum, at distinct hours.

07 Measure turbulence instead of assuming it

A burst of twenty successive frames on the same sightline. The standard deviation of the horizon position across those twenty frames IS the pointing term of the error budget. A grazing sightline crosses hundreds of kilometres of air near the surface — the geometry that produces the most optical turbulence — and a postulated figure would be optimistic there.

08 Measure the surface's unrest instead of assuming it

On the same burst of twenty frames, take the scatter of the *reflection's* position. A tilt ε of the surface displaces the reflection by 2ε and their midpoint by ε : **the scatter of the reflection divided by two is the**

surface's contribution to δ . It is measured, not postulated. Beyond 1', change liquid or shelter before going on.

09 Cross the two methods

One series with the artificial horizon AND one with the inclinometer and reversal, same site, same hour. The symmetry of the artificial horizon is an assumption until it is tested: the direct and reflected paths do not cross rigorously the same low layers. The gap between the two methods bounds that assumption.

WHY REFRACTION DOES NOT INTERFERE

The direct image of the star and its reflected image undergo the same refraction and remain symmetric about the horizontal. Their midpoint is therefore free of refraction, even though each of the two images is affected by it.

When to shoot — the observing window

This is method A's heaviest practical constraint, and no earlier version mentioned it: the frame must hold **the target, its reflection and the sea horizon line at once**. In full darkness the sea horizon line is no longer visible.

The useful window is therefore bounded at both ends: the target must have appeared, and the horizon must not yet have gone. It is narrow, but it comes round twice a day.

window	what is visible	duration
Sun between -5° and -10° — nautical twilight	stars of magnitude 1 to 2, and the horizon line still crisp	20 to 30 min, twice a day
Venus or Jupiter, before sunset	the target alone in a bright sky, and the horizon in full daylight	several hours
Moonlit night, Moon high	the horizon lit by the Moon; the Moon itself can serve as target	several hours

Table 5 — Practicable windows. Durations are for mid-latitudes; they shorten near the equator and lengthen, to the point of vanishing, at high latitudes in summer.

The most convenient target is therefore not a star: it is **Venus or Jupiter**. Bright enough to be acquired well before nightfall, when the sea horizon is still fully legible, and their reflection in the tray is found without trouble. The Moon serves too — the symmetry then applies to the centre of the disc.

CONSEQUENCE FOR PLATE SOLVING

In that window the field holds only a handful of stars — sometimes too few for a solver, which wants about a dozen. Take the measurement frame in the window, then a **second frame in full darkness, same setting, ring taped**, meant only for the solve. It is the scale that carries from one frame to the other, and it depends only on the optical setting — provided, precisely, that nothing was touched.

Where exactly the horizon line is

The horizon line is not a stroke: it is an intensity step spread over several pixels. At 100 mm on 4 000 pixels of height, one pixel is 0.21' and three pixels 0.62' — a quarter of the instrumental budget. Two operators pointing “by eye” are therefore not measuring the same quantity, and the discrepancy shows up nowhere in the data.

The criterion, to be fixed before reduction and applied without exception:

1. Work on the **raw, uncropped file**, with no automatic lens correction and no contrast enhancement.
2. For each pixel column, take the vertical intensity profile around the transition and read the ordinate of the **half-height point of the step**, by sub-pixel interpolation between the sea level and the sky level.

3. Average over the full usable width, after discarding columns containing a ship, an islet or a haze bank — and record how many columns were discarded.
4. Record the **column-to-column standard deviation** of that mean: it measures the regularity of the line, and an anomalous value flags a disturbed refractive regime before one even looks at the result.

There is nothing sacred about the half-height criterion. What matters is that it be **the same at every station**: a bias common to all shows up in the fitted intercept, a bias that varies between observers shows up nowhere. Any other reproducible criterion will do, provided it is stated in advance and applied throughout.

A free check on the same frame

The angular separation between the target and its reflection is exactly **twice its apparent altitude**. And the ephemeris gives its true altitude to the arcsecond. The comparison costs nothing and is worth making.

$$\text{measured separation} \div 2 = \text{true altitude} + \text{refraction}$$

At 3° altitude refraction is about 14.3'; at 4°, 11.7'; at 5°, 9.9'.

What this check establishes. It verifies the angular scale to about 0.5 per cent, independently of the solver — two routes that agree are worth more than one taken on trust. And it flags an atmosphere in an unusual state if the deduced refraction departs markedly from the standard value for the recorded pressure and temperature.

What it does not establish. It *does not measure* k along the grazing sightline. The path to a target at 3° crosses the whole troposphere obliquely, whereas the horizon line grazes the boundary layer over tens of kilometres: these are two distinct quantities, and conflating them would be exactly the kind of shortcut this document reproaches elsewhere.

Fallback — levelling and reversal

If an artificial horizon is impractical, measure with an inclinometer by the procedure below. It remains valid, but it depends on the repeatability of the station setup, where the artificial horizon does not.

01 Set up

Tripod on stable ground, base levelled in both axes. Pitch is the critical quantity; roll affects only framing.

02 Acquire the horizon, normal position

Focus at infinity, mid aperture, fast shutter. Save the raw file, uncropped.

03 Acquire the horizon, inverted position

Remount the camera upside down on the same base, re-level, reproduce the same framing.

04 Angular scale

Plate-solve a star field taken at the same setting, ring taped — section 08. **Do not trust the engraved focal length**, nominal to 2–5 per cent, nor the lunar standard, at 7 per cent.

05 Reduction

Measure in pixels the ordinate of the file centre and of the horizon line. Convert to angle using the sensor's vertical field of view.

06 Repetition

At least three series at distinct times, to sample refraction variability.

Eliminating mounting bias by reversal

The optical axis is generally not rigorously parallel to the camera base, and every inclinometer carries a zero error. These two defects combine into a pitch bias e undetectable in a single acquisition. In the inverted position e changes sign; the mean of the two readings eliminates it exactly.

$$\delta = (\delta_{\text{normal}} + \delta_{\text{inverted}}) / 2$$

A discrepancy greater than 20' between the two readings indicates a large mounting bias — the mean remains valid, but repeatability must be checked.

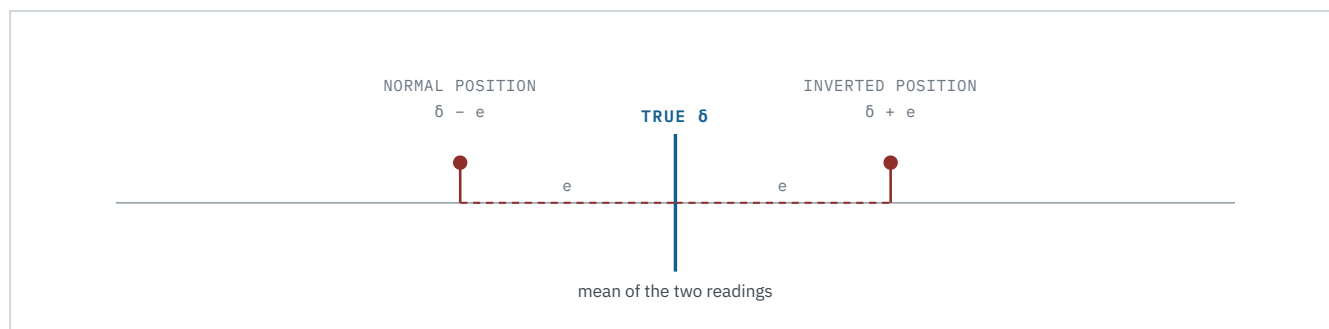


Figure 3 — Calibration by reversal. A standard surveying procedure for eliminating systematic instrumental bias, transposed here to the camera-inclinometer pair. It dispenses with knowing the bias, which is its principal advantage over any attempt at prior calibration.

07 Procedure — Method B, differential measurement

This variant requires **neither a levelling instrument nor a bias calibration** — the angular scale is still needed, as everywhere else. It consists of photographing the same horizon from two widely separated altitudes, with identical equipment and settings, and measuring the *displacement* of the horizon line within the frame. Any mounting bias, being identical in both acquisitions, cancels by difference.

RESTRICTION ON METHOD B

Cancellation of the bias assumes it is *identical* in both acquisitions. But going from 10 m to 3 107 m requires dismounting the camera, travelling, and remounting: the bias is no longer the same. A remounting repeatability of 20' consumes **21 per cent** of the 94' differential signal.

Method B remains useful when the camera-head assembly is not dismounted between the two stations. Otherwise use the artificial horizon of Method A, which depends on no mounting repeatability at all.

ALTITUDE PAIR	hypothesis S — k = 0.13	hypothesis P
2 m → 300 m	29'	0'
10 m → 1 000 m	51'	0'
50 m → 2 000 m	68'	0'
10 m → 3 107 m	94'	0'

Table 6 — Predicted displacement of the horizon line between two altitudes.

08 Error budget and decision criteria

Contributions are estimated at 1 σ for an acquisition at 100 mm on a 24 × 36 mm sensor sampled at 4 000 pixels vertically.

SOURCE	contribution	TREATMENT
Pointing the horizon line, turbulence included	to be MEASURED	scatter over a burst of 20 frames
Inclinometer zero ($\pm 0.05^\circ$)	3.0'	<i>eliminated</i> by reversal
Residual after reversal	1.5'	station re-setting repeatability
Optical distortion at field centre	< 1.0'	horizon placed near the axis
Definition of the horizon line	1.0'	three series, conditions recorded
Lunar standard ($\pm 7\%$ of diameter)	7.0'	<i>avoided</i> by the sensor method
Angular scale — nominal focal length to $\pm 3\%$	3.0'	eliminated by field astrometry (v1.3)
INSTRUMENTAL total	2.2'	sub-total
MODEL uncertainty — k varying with altitude	2.6'	model term
Local Gaussian radius of curvature ($\pm 0.3\%$ on R)	0.15'	negligible, noted for the record
Quadratic total	3.4'	1 σ

Table 7 — Error budget. The dominant term, levelling error, is eliminated by reversal; the residual is the repeatability of re-setting the station.

TWO SIGNAL-TO-NOISE RATIOS, NOT ONE

This protocol answers two distinct questions, and they do not call for the same ratio. Conflating them amounts to counting the uncertainty on k twice, and yields a needlessly pessimistic figure.

Question 1 — is the surface flat? The signal is the gap between $\delta = 0$ and the smallest dip a sphere can produce, that is **78.2'** at 3 107 m, reached at $k = 0.47$. The noise is the *instrumental* budget, 2.2'. Ignorance of k does not enter: it is already absorbed by taking the least favourable case, and no value of k brings a sphere closer to zero. **Ratio: 36.**

Question 2 — what is a? Here the measurement is set against a numerical prediction, and model uncertainty enters in full: $\sigma = 3.41'$ for $\delta = 100.1'$ at $k = 0.13$. **Ratio: 29.**

36 against the plane hypothesis | 29 on the determination of a

At 3 107 m. Table 11 gives both columns for every altitude.

Combining the least favourable signal with the widest noise would give 23. That figure counts k twice: the value of 78.2' *already* assumes the least favourable k in the whole range.

Model uncertainty

WHY A SINGLE K WILL NOT DO

Applying a **single k** at every altitude would be convenient. But k characterises the vertical density gradient, which differs sharply between the boundary layer — where 0.13 to 0.17 is well constrained — and the free troposphere above 1 000 m, where it is lower. From 3 107 m the path crosses both.

k	δ	regime
0.08	103.0'	free troposphere
0.10	101.8'	high altitude
0.13	100.1'	geodetic standard
0.15	99.0'	boundary layer
0.17	97.8'	cold sea

Table 8 — Sensitivity of δ to the refraction coefficient. The spread over a plausible range exceeds the entire instrumental budget.

Spread: **5.2'**, that is $\pm 2.6'$. That exceeds the instrumental budget of 2.2', and the term is real: it bears on the *prediction*, and therefore on the determination of a.

WHICH QUESTION THIS TERM BELONGS TO

It must not be charged against the discrimination between the two models. That rests on the gap between $\delta = 0$ and 78.2', a value that *already* assumes the least favourable k in the whole range: adding $\pm 2.6'$ of k spread on top would count k a second time.

The term belongs to the second question — what is a? — where the measurement is set against a numerical prediction. The two ratios are given separately above.

The proper refinement is an **altitude-dependent refraction model** rather than a single k. An observer holding a temperature profile along the sightline — sounding or reanalysis — must use it and record it.

The angular scale

A TERM EASILY OVERLOOKED

δ is read as an offset in pixels, converted to an angle by the focal length and the pixel pitch. But **the focal length engraved on a lens is nominal**: the true focal length at infinity commonly departs from it by 2 to 5 per cent, and focus breathing varies it further. A relative error on the scale carries over in full to δ . The lunar standard is set aside — ± 7 per cent of the diameter, that is 7.0' — but the “sensor method” that replaces it carries a term of its own, and that term must be quantified.

angular scale	term induced	total at 1 σ	ratio on the law
field astrometry	0.05'	3.41'	29.4
nominal focal length to $\pm 2\%$	2.0'	3.95'	25.3
nominal focal length to $\pm 3\%$	3.0'	4.54'	22.1
nominal focal length to $\pm 5\%$	5.0'	6.05'	16.5

Table 9 — What a focal length taken at face value costs, at 3 107 m where δ is 100'. Ratios given in the “on the law” convention — convention.

The “against $\delta = 0$ ” ratio does not appear in this table, and deliberately so: a scale error is **multiplicative**. It cannot manufacture 78' out of zero, so it does not threaten discrimination between the two models. It falsifies the *value* of a, and that is where it must be counted.

The clean solution exists and it is free: **field astrometry**, or plate solving.

1. On the frame itself, or on one taken at the same setting without touching the ring, solve the star field — Astrometry.net, ASTAP, SIRIL, PixInsight, all free or open.

- The solver returns the **scale in arcseconds per pixel**, fitted on dozens of catalogue stars, with its residuals. Ordinary accuracy is of order 0.05 per cent — that is **0.05' in 100'**, a thousand times better than the nominal focal length and a hundred times better than the lunar standard.
- It also returns the field's **distortion profile** (SIP terms), which disposes of the “optical distortion” line of the error table at the same stroke.

Method A already puts a star and its reflection in the field: the acquisition is nocturnal, the field is populated, the solve is immediate. For a daytime frame, photograph a star field the same evening at the identical setting, ring taped. **The solved scale and the solver's residuals become mandatory fields on the record sheet:** an angular measurement whose scale is undocumented has no estimable uncertainty.

Decision criteria, fixed a priori

δ MEASURED $\in [0 ; 20']$ AT 3 107 m

Hypothesis S refuted

No value of k within the physical range falls below 78'. A measurement in this interval, reproduced over three series, refutes the spherical model by direct measurement.

δ MEASURED $\in [78' ; 107']$ AT 3 107 m

Hypothesis P refuted

The exact position within the interval additionally provides an independent estimate of the refraction coefficient at the time of observation.

DEAD ZONE — 20' TO 78'

Neither hypothesis predicts this range. A result falling within it must be treated as a **protocol failure**, not a physical result. Causes to rule out, in order of frequency: faulty levelling in pitch, false horizon line (coastline, island, haze bank), cropping of the file, focal length changed between acquisitions.

What each altitude is really worth

Signal-to-noise collapses at low stations. Those points remain useful for fitting the $\sqrt{h}$ law, but they are not decisive and must not be presented as such.

altitude	δ ($k = 0.13$)	total σ	against $\delta = 0$	on the law	use
5 m	4.0'	2.20'	1.4	1.8	indicative only
10 m	5.7'	2.20'	2.0	2.6	indicative only
50 m	12.7'	2.22'	4.5	5.7	contributory
120 m	19.7'	2.26'	7.0	8.7	contributory
300 m	31.1'	2.34'	11.0	13.3	solid
1 000 m	56.8'	2.65'	20.2	21.4	solid
3 107 m	100.1'	3.41'	35.5	29.4	decisive

Table 10 — What each altitude is worth, for each of the two questions. The “against $\delta = 0$ ” column takes the least favourable signal ($1.4022\sqrt{h}$, $k = 0.47$) over the instrumental budget alone; the “on the law” column takes δ at $k = 0.13$ over the full budget.

The two columns cross near 1 000 m, and the reason is worth seeing: below that altitude model uncertainty is negligible against the instrumental budget and it is the margin on k that weighs, while above it model uncertainty grows with δ whereas the worst case on k stays proportional to $\sqrt{h}$. **In short: a high station is better for refuting the plane, and a middling station costs hardly more for measuring a.**

The elevation datum — the most dangerous point

THE DATUM, AND WHY IT IS DANGEROUS

δ depends on height above the **sea surface at the time**. Writing “ground elevation + tripod height” without specifying the datum will not do: a GPS receiver returns an *ellipsoidal* height, which may depart from sea level by tens of metres.

Consequence, for a 30 m datum error: **178 per cent error on δ at 10 m**, 63 per cent at 50 m, 10 per cent at 300 m, 1 per cent at 3 107 m. At low stations it destroys the measurement. Require **orthometric** elevation, and below 50 m a direct measurement above the water.

altitude	tide ± 2 m	swell $H_s = 3$ m
5 m	41 %	—
10 m	20 %	−7.8 %
50 m	4.0 %	−1.5 %
300 m	0.7 %	−0.3 %
3 107 m	< 0.1 %	< 0.1 %

Table 11 — Effects of tide and sea state. The second is a systematic bias, always of the same sign.

Swell does not add noise: it **lowers** δ , because the crests mask the mean surface. At 10 m in a formed sea the bias is 0.4', of the same order as the entire error budget. Tide and sea state are now mandatory fields of the record.

09 Known systematics and failure modes

1. **False horizon line.** The most frequent and most insidious failure mode. A distant coastline, an island, a mountain range or a haze bank produces a visual limit that is not the geometric horizon. Cartographic verification before occupying the station is mandatory.
2. **Pitch confused with roll.** A horizon rendered level in the image says nothing about the vertical tilt of the optical axis, the only quantity measured here.
3. **Cropping.** Any cropped image loses the position of its optical centre and therefore its horizontal reference. Work exclusively on the native file.
4. **Optical distortion.** Wide-angle lenses curve the horizon off axis. Place the horizon line near the centre, or use a telephoto lens.
5. **Focal drift.** Focal length must be identical between the measurement frame and the frame meant for plate solving — the whole validity of carrying the scale across rests on it. Tape the ring, and beware internal-focus lenses, whose focal length varies with distance.
6. **Anomalous refraction regime.** A horizon that undulates, doubles, or shows a sharp cut-off line indicates an atmospheric duct. The measurement remains valid but the value of k it implies is not transferable to other observations. Always record the appearance of the horizon.
7. **Single measurement.** An isolated acquisition has no estimable uncertainty. Three series are the minimum.

The boundary layer over water

The sensitivity table of section 08 deals with k varying *with altitude*. A second effect remains, and it strikes low stations precisely: **the difference between air and water temperature**. It sets the thermal gradient of the first few metres, and the gradient sets k .

dT / dh	k	regime
−0.30 K/m	−1.63	strongly unstable, cold air over warm water — inferior mirage
−0.05 K/m	−0.10	unstable
−0.013 K/m	0.13	geodetic standard
0.00 (isothermal)	0.21	neutral
+0.05 K/m	0.52	inversion — superior mirage
+0.30 K/m	2.05	atmospheric duct

Table 12 — Refraction coefficient by vertical thermal gradient, from $k = 503 (P/T^2)(0.0342 + dT/dh)$, at 1 013 hPa and 288 K. Pressure is in hectopascals: in pascals the coefficient comes out a hundred times larger, and that is a classic trap.

These values assume the gradient holds along the whole sightline. In practice the affected layer is only metres thick — which is exactly why it dominates below 50 m, where the sightline never leaves it, and becomes negligible from a summit. It also accounts for disagreement between two readings at the same site at different hours, where nothing instrumental has changed.

WHAT THIS IMPOSES ON THE RECORD SHEET

Record the **sea surface temperature** and the air temperature at eye height, and log their difference $\Delta T = T_{\text{air}} - T_{\text{sea}}$. A reading without ΔT does not characterise its own regime.

$\Delta T > +2$ °C (colder water): stable layer, inversion, duct possible. The horizon looks distant and too sharp, and δ is *smaller* than the standard prediction.

$\Delta T < -1$ °C (warmer water): unstable layer, inferior mirage. The horizon looks close and boiling, and δ is *larger*.

$|\Delta T| < 1$ °C: near-standard regime. These are the most useful stations, and they should be sought rather than awaited.

Stations at $|\Delta T| > 2$ °C are **kept, published and flagged**, then analysed separately. None is deleted: that is the rule set in section 09, and it admits no exception.

The direction of these effects is worth stressing, because it runs against common intuition: it is *cold* air over warm water that brings the horizon closer and deepens the dip, and warm air over cold water that pushes it away, to the point of raising objects reputed to be hidden. A spectacularly long-range observation is therefore not an observation in the ordinary regime, and ΔT says so.

10 Distributed campaign — what we propose

The value of a collective campaign goes beyond repeating a single measurement. Plotting δ against $\sqrt{h}$ turns the question into **three independent constraints**, each separately verifiable.

1. **The slope.** The fitted coefficient a must fall within [1.40; 1.93] for hypothesis S, or be consistent with zero for hypothesis P.
2. **Linearity.** Points must align in δ against $\sqrt{h}$. A relation of any other form would refute both hypotheses and indicate an unidentified systematic.
3. **Passage through the origin.** The fitted intercept must be consistent with zero. A significantly non-zero value would reveal a bias common to all observers.

We therefore invite observers at staggered altitudes — from sea level to high-altitude stations — to contribute one point each. About ten points spread over the range 5 m – 3 000 m suffice to constrain the slope to better than 5 per cent.

Fitting method

The three constraints above are tested thus:

- **Weighted regression** of δ against $\sqrt{h}$, weights being the inverse square of each station's uncertainty — the signal-to-noise table by altitude already supplies them. Unweighted, a station at 5 m counts as much as one at 3 000 m, which is absurd.
- **Confidence intervals** on the slope a AND on the intercept, not just points. A slope of 1.7 with no error bar says nothing.
- **Randomised order**: alternate normal and inverted in a drawn order, and vary azimuths. Without this, a slow refraction drift through the session is indistinguishable from an instrumental bias.

The free exponent

The three constraints above assume the form $\delta = a\sqrt{h}$. That grants in advance what the measurement ought to establish. The general form $\delta = a \cdot h^p$, with the exponent p left free, tests it instead of postulating it.

- The sphere predicts $p = 0.500$ exactly, with no adjustable parameter.
- The plane predicts $a = 0$, and p then loses all meaning.
- A law linear in h — what one would get if δ were in fact measuring a bias proportional to the height of the mounting — would give $p = 1$.

The fit is carried out in logarithms, $\ln \delta = \ln a + p \ln h$, weighted by $(\delta/\sigma_\delta)^2$. With six stations staggered from 10 m to 3 107 m, one series each and the error budget of section 08:

WHAT THE EXPONENT CAN SETTLE

$\sigma_p = 0.022$, and 0.013 with three series per station. The gap between $p = 0.5$ and $p = 1$ is therefore **22 σ** ; even $p = 0.6$ would be separated from $p = 0.5$ at 4.5 σ . The exponent is measurable. It need not be assumed, and assuming it was the last concession this protocol made to the hypothesis it claims to test.

Two precautions. First, the logarithmic fit forces $b = 0$: the linear fit $\delta = a\sqrt{h} + b$ must therefore be run *as well*, and b checked for consistency with zero. If b is not consistent with zero, the exponent drawn from the logarithm is contaminated by that bias, and both fits must be redone together, on three parameters.

Second, k varying with altitude — the sensitivity table — biases the exponent upward. If low stations work at $k = 0.15$ and the summit at $k = 0.10$, δ is 2.9 per cent larger there, which adds **+0.005** to p . Five times under the statistical uncertainty, but to be logged rather than discovered afterwards.

CONDITIONS OF PARTICIPATION

Publication of **raw, uncropped files**, complete EXIF metadata, the observation record, and the result **whatever it is**. No data discarded without written justification. Acquisitions indicating an anomalous refractive regime are retained and flagged, not deleted.

11 Data to be submitted

Date and UTC time (start / end of series)	_____
GPS coordinates of the station	_____
Eye height above the sea surface at the time (orthometric)	_____
How that height was determined, and its uncertainty	_____
Tide time and range	_____
Sea state, significant wave height H_s	_____
Method used: artificial horizon, or level with reversal	_____
Viewing azimuth, and open-sea distance along that axis	_____
Temperature, pressure, humidity, wind	_____
Camera, lens, exact focal length, sensor dimensions	_____
Vertical pixel count of the native image	_____
Inclinometer used, manufacturer's accuracy	_____
Normal position — horizon ordinate (px)	_____
Inverted position — horizon ordinate (px)	_____
Ordinate of the file centre (px)	_____
Reduced δ (arcminutes) and estimated uncertainty	_____
Horizon appearance: sharp / undulating / doubled / cut-off line	_____
Number of series and observed scatter	_____
Air temperature at eye height, and sea surface temperature	_____
$\Delta T = T_{\text{air}} - T_{\text{sea}}$, and the regime it indicates	_____
Liquid used for the surface, and settling time before the frame	_____
Angular scale solved by astrometry ("/px), and solver residuals	_____
Scatter of the reflection over the burst ('), and the half retained	_____
Window used, and target sighted (name, ephemeris altitude)	_____
Measured target-to-reflection separation, and the refraction deduced	_____
Horizon pointing criterion, and column-to-column standard deviation	_____
Columns discarded from the mean, and why	_____

12 Commitment

This protocol was written before any data were acquired, and states in advance the meaning of each of the two outcomes. That is the condition of its evidential value.

WHAT IS STILL MISSING, AND MUST BE DONE BEFORE COLLECTING

A version number proves nothing. “Version 1.9” is not opposable to a sceptical reader: nothing materially prevents rewriting a document after seeing the data.

DOI of this deposit: 10.5281/zenodo.22167798

While this line reads “pending”, the pre-registration is not binding, and the document says so itself.

A **timestamp verifiable by a third party** is required. The procedure, in this order, so that the DOI appears in the document itself rather than beside it:

1. **Open the deposit without publishing it.** On Zenodo, create the record and request *Reserve DOI*: the DOI is assigned before publication. OSF offers the same.
2. **Write the reserved DOI** into the two slots provided — front page and above — then regenerate the French and English PDFs.
3. **Freeze.** No modification afterwards, not even a typo: any retouching invalidates the hashes.
4. **Compute the SHA-256 hash** of each of the two frozen files.
5. **Upload both files to the SAME record** — the DOI bears on the record, not on each file — paste the hashes into the notes field, and publish.
6. **Only then begin collecting.**

WHY THE DOI CAN SIT INSIDE THE FILE AND THE HASH CANNOT

The DOI is assigned *before* publication: it can therefore enter the document, which is then frozen and hashed. The hash, by contrast, is computed *on* the finished file — writing it in would alter the file, and so falsify it. It necessarily lives outside: in the record's notes, and wherever the protocol is announced.

The DOI supplies the date, the hash supplies the file's exact identity; either alone can be worked around, the two together make the pre-registration binding.

A point of method, to forestall a legitimate objection. Writing in the DOI does not change the version number: it fills a slot opened for the purpose. It follows that version 1.8 exists publicly only in its form *bearing the DOI*. The copy with an empty slot is an intermediate working state and must not circulate: two distinct files claiming the same version number would be exactly the kind of ambiguity this arrangement exists to remove.

History. This document was revised several times before deposit, following outside review. No earlier version was deposited or circulated: the full revision history is kept with the sources and supplied on request. It is not reproduced here, where it would teach nothing to a reader who has come to carry out the protocol.

We undertake to publish the entirety of the data received, including those contradicting the hypothesis we judged more probable, and to publish the conclusion the data impose. A protocol whose result were known in advance would not be worth executing; this one is proposed precisely because its outcome is not.

Methodological comments, corrections and proposals for improving the protocol are invited before the campaign begins, and will be incorporated into a credited version 2.

13 References

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All numerical values in this document derive from $\delta = \sqrt{2h/R'}$ with $R' = R/(1-k)$ and $R = 6\,371$ km. Mean lunar diameter $31'$ (29.4' at perigee, 33.5' at apogee). The document is reproducible in its entirety from these relations. — Open protocol, version 1.9.